Dream Analysis and Prediction

PROJECT REPORT

Submitted in partial fulfilment of the requirements
for the award of the degree of

BACHLOR'S OF COMPUTER APPLICATIONS

Submitted by

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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the PROJECT REPORT entitled as Dream Analysis project for semester 6th in partial fulfillment of the requirements for the award of the Degree of Bachelor of Computer

Applications AI&DS and submitted in the School of Computing of the Graphic Era Hill

University, Dehradun is an authentic record of my own work carried out under the supervision of MS.Nidh Joshi .

Signature of the Candidate

Signature of Mentor (s)

Acknowledgement

I would like to express my sincere gratitude to everyone who supported me in completing this project. First and foremost, I thank my teacher/mentor for their valuable guidance, encouragement, and continuous support throughout the development of this project.

I am also thankful to my college for providing the necessary resources and a learning environment that helped me work on this project effectively. I would like to acknowledge my friends and classmates for their suggestions, motivation, and help whenever I faced difficulties.

Lastly, I would like to thank my family for their constant support and understanding, which kept me motivated during the entire process.

This project has been a great learning experience for me, and I am truly grateful to everyone who contributed directly or indirectly to its completion.

Signature of candidate

Abstract

This project presents an intelligent web-based application that analyzes user-described dreams and generates meaningful interpretations using Machine Learning and symbolic analysis. The system combines Natural Language Processing (NLP) with a supervised learning model to classify dreams into categories and emotional moods. Additionally, it enhances interpretation by incorporating astrological, spiritual, and psychological perspectives.

The application is built using **Flask (backend)**, **HTML (frontend)**, and **Scikit-learn (ML model)**. A dataset of dream descriptions is used to train the model, enabling it to predict dream categories and moods with reasonable accuracy. The system also extracts key symbols from the dream text and provides multi-dimensional insights.

Introduction

Dreams have always been a subject of curiosity and interpretation across psychology, spirituality, and astrology. Traditionally, dream analysis required human experts, but with advancements in Artificial Intelligence, it is now possible to automate this process.

This project aims to develop an AI-powered system that:

- Accepts a user’s dream description
- Analyzes it using Machine Learning
- Provides meaningful interpretations

The system not only predicts the dream type but also explains it from:

- Psychological perspective
- Spiritual meaning
- Astrological influence

Motivation

The motivation behind this project includes:

- To apply Machine Learning in a unique domain (dream analysis)
- To create an interactive and user-friendly application
- To combine technical AI with human-centric interpretation
- To explore how NLP can understand human emotions and experiences

Project Description

Dream Sense is an AI-powered dream analysis and prediction system that takes a user's dream description as input and generates interpretations from three unique perspectives:

Astrological - based on zodiac sign and celestial symbolism

Spiritual - based on sacred symbols and soul-level meanings

Psychological - based on Jungian psychology and emotional analysis

The system uses a Machine Learning model (Random Forest + TF-IDF) trained on a dataset of 500 dreams to predict:

The dream category (Spiritual, Fear, Transformation, Career, etc.)

The dream mood (Anxious, Peaceful, Joyful, etc.)

Probability breakdown of top categories

Problem Statement

Traditional dream interpretation is:

- Subjective
- Time-consuming
- Dependent on human experts

There is a need for an automated system that:

- Quickly analyzes dream descriptions
- Provides structured interpretations
- Works consistently for all users

Objectives

- To build a web-based dream analysis system

- To classify dreams into categories using ML
- To detect emotional mood from dream text
- To extract symbolic meanings
- To provide multi-layered interpretation

Technologies Used

- Python
- Flask (Web Framework)
- Scikit-learn (Machine Learning)
- Pandas & NumPy (Data Processing)
- HTML/CSS (Frontend)

Dataset Description

The dataset (Dream.csv) contains:

- dream_description → Text input
- dream_category → Label (e.g., Fear, Career, etc.)
- dream_mood → Emotional state

Preprocessing Steps:

- Removed missing values
- Encoded categorical labels using **LabelEncoder**
- Converted text into numerical form using **TF-IDF**

Methodology

1. Text Processing

The system uses **TF-IDF Vectorization**:

- Converts text into numerical vectors
- Focuses on important words
- Removes common stopwords

2. Machine Learning Model Model Used:

Random Forest Classifier

Two separate models are trained:

- One for **Dream Category**
- One for **Dream Mood**

Why Random Forest?

- Handles text features well
- Reduces overfitting
- Provides good accuracy

3. Model Training Process

1. Load dataset
2. Clean data
3. Encode labels
4. Apply TF-IDF
5. Train classifiers

4. Prediction Flow

When user enters a dream:

1. Text → TF-IDF vector
2. Model predicts:
    - Category
    - Mood
3. Extract symbols from text
4. Generate interpretations

System Architecture

User Input (DreamText)
↓
Flask Backend
↓
TF-IDF Vectorization
↓
Random ForestModel
↓
Prediction (Category+Mood)
↓
Symbol Extraction
↓
Final Interpretation Output

Code Explanation

1. Model Training Function

def train_model():

- Reads dataset
- Applies preprocessing
- Trains two classifiers

2. Symbol Extraction

def extract_symbols(text):

- Searches keywords like:
    - water, snake, fire, etc.
- Returns top 5 symbols

3. Prediction Function def predict_dream(description, zodiac, age):

This is the core logic:

- Converts input text
- Predicts category & mood
- Calculates probabilities
- Generates interpretations

4. Flask API

@app.route('/analyze', methods=['POST'])

- Receives user input
- Calls prediction function
- Returns JSON response

Key Features

- Dream classification using ML
- Emotion detection
- Symbol-based interpretation
- Zodiac-based insights
- Age-based psychological analysis

Result & Output

The system outputs:

- Predicted Dream Category
- Predicted Mood
- Top 3 probable categories
- Extracted symbols
- Interpretations: ○ Astrological ○ Spiritual ○ Psychological

Advantages

- Automated dream analysis
- Multi-perspective interpretation
- Fast and user-friendly
- Scalable system

Limitations

- Accuracy depends on dataset quality
- Limited symbol dictionary
- Interpretations are generalized

Future Scope

- Use Deep Learning (LSTM, BERT)
- Expand dataset
- Add voice input
- Improve personalization
- Mobile application development

Conclusion

The project successfully demonstrates how Machine Learning and NLP can be used to analyze human dreams. By combining technical models with symbolic interpretation, the system provides a unique and meaningful output. It shows the potential of AI in understanding human emotions and experiences beyond traditional applications.

References

- Scikit-learn Documentation
- Flask Documentation
- NLP (TF-IDF concepts)
- Dream psychology theories (Freud & Jung)

DREAMSENSE - DREAM ANALYSIS AND PREDICTION SYSTEM PSEUDOCODE

START

INITIALIZE Flask web application

LOAD Dream dataset (Dream.csv)

CLEAN dataset:

REMOVE rows with missing dream description, category, or mood

ENCODE labels:

CONVERT dream categories and moods into numerical values using LabelEncoder

FEATURE EXTRACTION:

APPLY TF-IDF vectorizer to convert dream text into numerical feature vectors

MODEL TRAINING:

TRAIN Random Forest Classifier for:

1. Dream Category prediction
2. Dream Mood prediction

STORE trained models and encoders

DEFINE function extract_symbols(text):

CONVERT text to lowercase

SEARCH for predefined symbols in text

RETURN top 5 matched symbols

```
DEFINE function predict_dream(description, zodiac, age):

   TRANSFORM description using TF-IDF

   PREDICT    category    and    mood    using    trained    models
CALCULATE prediction probabilities

   SELECT top 3 categories with highest probabilities
   EXTRACT symbols from description

   GENERATE interpretations:

- Astrological meaning

- Spiritual meaning

- Psychological meaning

   RETURN prediction results


DEFINE route "/":

   DISPLAY homepage (index.html)


DEFINE route "/analyze":

   RECEIVE user input (dream description, zodiac, age)

   IF description length < 20:
RETURN error message     ELSE:

      CALL predict_dream()

      RETURN results in JSON format


RUN Flask server on port 5000


    END
```